

L4.1b Functions of y

Sideways rectangles — what you need to know

The swap

The input runs UP the page. The output runs ACROSS it. Every row below is a consequence of that one line.

You know this for $y = f(x)$

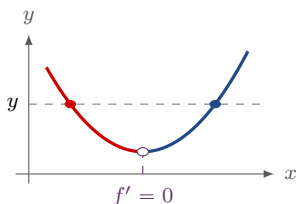
slope = m : rise over run
 quadratic, $a > 0$ opens up; $a < 0$ opens down
 the zeros are x -intercepts
 $f(0)$ is the y -intercept
 Loh's midline m : the vertical line $x = m$
 denominator = 0: vertical asymptote
 limit at $\pm\infty$: horizontal asymptote
 vertical line test

Same idea for $x = g(y)$

$m = \Delta x / \Delta y$: **run over rise**
 opens **right**; opens **left**
 the zeros are y -intercepts
 $g(0)$ is the x -intercept
 the **horizontal** line $y = m$; axis of symmetry horizontal
horizontal asymptote
vertical asymptote
horizontal line test

Transformations are unchanged. Outside affects the output normally; inside affects the input backward. Outside now moves x , across. Inside still runs backward.

MONOTONIC PIECES

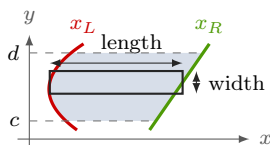


$x = f^{-1}(y)$ exists on any interval where f is monotonic

“every horizontal line crosses once, so each y comes from one x ”

Not monotonic on the interval? **Split at $f'(x) = 0$.** On each piece it is, and each piece solves for x — the $\pm$ branches are the pieces.

AREA WITH SIDEWAYS RECTANGLES



$$A = \int_c^d x_R - x_L dy$$

both boundaries continuous on $[c, d]$, written as x in terms of y

$$\text{length} = x_R - x_L, \quad \text{width} = \Delta y, \quad A = \lim_{n \rightarrow \infty} \sum_{i=1}^n (x_R - x_L) \Delta y$$

“each rectangle is as long as the gap between the curves and as tall as Δy ”

- ① **Solve each boundary for x .**
- ② **Get the limits.** Solve $x_R = x_L$ — they are the y -coordinates of the crossings.
- ③ **Test a value of y** to decide which boundary is on the right. Evaluate both there and compare.
- ④ **Integrate right minus left.** If the curves trade places, one integral per piece.

WHICH RECTANGLE

Draw one. If both its ends stay on one curve each all the way across the region, that is your variable. Upright: $y_T - y_B dx$. Sideways: $x_R - x_L dy$.

Two reasons to split, and they are different.

A boundary turns around in the interval	x_L and x_R cross in the interval
a <i>graphing</i> problem — split to <i>write</i> it as $x =$	an <i>area</i> problem — split the <i>integral</i> : right
$g(y)$, one branch per piece. The area may still	and left trade places, exactly as top and bot-
be one integral	tom did

“split at the crossing, not at the turn”

WATCH OUT

✗ $g(0)$ is the *y*-intercept → the *x*-intercept — input 0 is the horizontal axis, and the output lands on it

✗ the slope of $x = 2y + 1$ is really $\frac{1}{2}$ → read it as $\Delta x / \Delta y = 2$: up 1, across 2

✗ ~~turn the page to make the graph upright~~ → nothing was reflected — you changed which variable you chose

✗ ~~the limits are the x -coordinates~~ → the y -coordinates — you are integrating dy

✗ ~~not monotonic, so it cannot be integrated in y~~ → the integral only needs every horizontal line to cross the region once

✗ ~~split at the turning points~~ → split at the crossings, where x_L and x_R trade places